Purchase & Sales Sub-Ledgers — Deep HLD Design

> ***Document Focus:*** *Accounts Payable (AP) & Accounts Receivable (AR) Architecture · Exact GL Mapping Diagrams · Real-World Examples*
> ***Project:*** *GTR Service ERP — Multi-Tenant SaaS Platform*
> ***Reference:*** *Accounting Module HLD · Accounting Module (3).docx*
> ***Style:*** *HLD Design Only — No Code*

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1. What are Sub-Ledgers?

In the GTR Service ERP, **Sub-Ledgers** handle the operational, day-to-day transactions with external business partners—specifically **Suppliers** (Purchases/AP) and **Customers** (Sales/AR).

Real-World Analogy:
Think of the General Ledger (GL) as the vault manager at a bank, and the Sub-Ledgers as the bank tellers. The tellers (AP/AR modules) interact with the customers, handle the specific paperwork (invoices, receipts), and understand the business context. At the end of the transaction, the teller hands a perfectly balanced summary (Journal Entry) to the vault manager (GL Engine) to officially record the movement of money.

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2. Sub-Ledger Architecture Mind Map

```
mindmap
  root((Sub-Ledgers Engine))
    Purchases (AP)
      Supplier Invoices
        Inventory / Goods
        Admin Expenses
        Input VAT
      Payments Made
        Bank Transfer
        Cash on Hand
        PDC Issued
      Prepayments to Suppliers
      Debit Notes & Corrections
      Returns to Suppliers
    Sales (AR)
      Customer Invoices
        Sales of Goods / Services
        Output VAT
        Discounts & Rebates
      Receipts (Payments In)
        Bank Transfer
        Cash on Hand
        PDC Received
      Prepayments from Customers
      Credit Notes & Corrections
      Customer Returns
    Integration Bridges
      Company GL (Partner Default)
      Item GL (Product Default)
      VAT Engine (Tax Rules)
```

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3. Dynamic GL Connection (How They Connect)

The Sub-Ledgers do **not** have hardcoded GL accounts. For example, the system never assumes that "Receivables" is always GL Code `3010001`. Instead, they connect dynamically to the Chart of Accounts (COA) using the **Three-Bridge System**.

Real-World Example: TechCorp buying from Dell

Imagine **TechCorp** (our tenant) purchases laptops from **Dell** (a Supplier) to resell. When the procurement officer clicks "Post" on the Purchase Invoice, the system runs this logic:

1. **Who is the partner?** (Bridge 2: Company GL)
    - The system looks up "Dell's" `CompanyGLAccount` profile and finds `accountsPayable` = `2001001 - Trade Payables (IT Suppliers)`.
2. **What is being bought/sold?** (Bridge 1: Item GL)
    - The system looks up the `ItemGLAccount` for the item "Dell XPS 15" and finds `goodsAccount` = `1002001 - Electronic Inventory`.
3. **Is VAT applicable?** (VAT Engine)
    - The invoice indicates 5% VAT. The VAT engine pulls the `Current VAT Receivable` GL.

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4. Purchase Sub-Ledger (Accounts Payable) — Deep Dive

4.1 Purchase Invoice Received — GL Mapping

When a **Purchase Invoice** is received and posted, the system auto-generates:

```
┌──────────────────────────────────────────────────────────────┐
│                  INVOICE RECEIVED (AP)                       │
├──────────────────┬───────────────────────────────────────────┤
│  Debit (Dr)      │  Credit (Cr)                              │
├──────────────────┼───────────────────────────────────────────┤
│ Goods (Inventory │ Current Trade Payables (Supplier balance) │
│  account)        │                                           │
│ Input VAT        │ Current Trade Payables (VAT portion)      │
│ Admin Expenses   │ Current Trade Payables (Expense portion)  │
└──────────────────┴───────────────────────────────────────────┘
```

> ***Real-World Explanation:***
> *We buy **1,000 kg of Steel Wire** from Emirates Steel LLC for AED 10,000 + AED 500 VAT.*
>
> - ***Dr Goods (Inventory)**: AED 10,000 (Asset increases)*

- ***Dr Input VAT**: AED 500 (Tax asset increases)*
- ***Cr Current Trade Payables**: AED 10,500 (Liability to Emirates Steel increases)*

4.2 Purchase Payment — GL Mapping

Payment Method	Dr	Cr
Bank Transfer	Current Trade Payables	Bank Account
Cash	Current Trade Payables	Cash on Hand
Cheque (PDC)	Current Trade Payables	PDCs Issued to Companies

***Bank charges on PDC:** Additional entry Dr Bank Charges / Cr Bank Account*

Real-World Explanation:
Two weeks later, we pay Emirates Steel LLC the AED 10,500 via Bank Transfer.

- ***Dr Current Trade Payables**: AED 10,500 (Liability drops to zero)*
- ***Cr Bank Account**: AED 10,500 (Our cash asset decreases)*

4.3 Supplier Prepayment — GL Mapping

PREPAYMENT TO SUPPLIER	
Dr	Cr
Prepayments to Suppliers	Bank / Cash on Hand / PDCs Issued to Companies

Real-World Explanation:
We wire an advance of AED 5,000 to German Tech GmbH before they

start manufacturing our custom machine.

- ***Dr Prepayments to Suppliers**: AED 5,000 (This is an Asset because they technically owe us that value until the machine arrives).*
- ***Cr Bank Account**: AED 5,000 (Cash decreases).*

4.4 Return of Goods to Supplier — GL Mapping

RETURN OF GOODS (SUPPLIER)	
Dr	Cr
Prepayments to Suppliers	Goods (Inventory)
	COS: Goods for Resales
	Current VAT Payable
Current Trade Payables	Prepayment from Customer

Real-World Explanation:
*A complex reversal if we return goods back to the supplier. If we haven't paid them yet, we reduce **Current Trade Payables** (Dr). If we already paid them, we record it as a **Prepayment to Suppliers** (Dr) because they now owe us a refund or credit. The Goods inventory is credited (reduced), and related VAT is reversed.*

4.5 Debit Note / Invoice Correction — GL Mapping

Event	Dr	Cr
Debit Note	Goods	Prepayments to Suppliers
	Prepayment to Supplier	Current Trade Payables
Mutual Settlement	Current Trade Payables	Prepayment to Supplier

Event	Dr	Cr

> ***Real-World Explanation:***
> *We discover Emirates Steel overcharged us by AED 500 on an invoice we already paid. They issue us a Credit Note (which we record as a Debit Note).*
>
> - ***Dr Prepayment to Supplier****: AED 500 (Creating a credit balance they owe us).*
> - ***Cr Current Trade Payables****: AED 500 (Adjusting the ledger).*

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5. Sales Sub-Ledger (Accounts Receivable) — Deep Dive

5.1 Customer Invoice — GL Mapping

CUSTOMER INVOICE (AR)	
Debit (Dr)	**Credit (Cr)**
Trade Receivables (full invoice)	Sales of Goods
Trade Receivables (VAT portion)	Current VAT Payable (Output VAT)
Trade Discount & Rebates	Sales of Goods (contra for discount)

> ***Real-World Explanation:***
> *We sell **5,000 Plastic Hangers** to Zara Fashion Retail for AED 2,000 + AED 100 VAT. We also give them a AED 100 discount.*
>
> - ***Dr Trade Receivables****: AED 2,100 (Asset: what Zara owes us).*
> - ***Dr Trade Discount & Rebates****: AED 100 (Expense: the discount given).*
> - ***Cr Sales of Goods****: AED 2,100 (Income: Gross value of goods).*

- ***Cr Current VAT Payable****: AED 100 (Liability: We owe this tax to the government).*

5.2 Payment Receipt — GL Mapping

Payment Method	Dr	Cr
Bank Transfer	Bank Account	Current Trade Receivables
Cash	Cash on Hand	Current Trade Receivables
Cheque (PDC)	PDCs Received from Companies	Current Trade Receivables

> ***Real-World Explanation:***
> *Zara sends a cheque for AED 2,100.*
>
> - ***Dr PDCs Received****: AED 2,100 (Asset: physical cheque in hand).*
> - ***Cr Current Trade Receivables****: AED 2,100 (Zara's balance drops to zero).*

5.3 Customer Prepayment — GL Mapping

```
Prepayment from Customer:
  Dr: Cash / Bank / PDC Received
  Cr: Prepayment from Customer (Liability account)
```

> ***Real-World Explanation:***
> *A new client, Boutique Inc., pays a 50% upfront deposit of AED 5,000 before we start production.*
>
> - ***Dr Bank Account****: AED 5,000 (Cash goes up).*
> - ***Cr Prepayment from Customer****: AED 5,000 (Liability: We owe them the products, or we owe them their money back if we fail).*

5.4 Credit Note / Invoice Correction — GL Mapping

CREDIT NOTE (AR — CUSTOMER)	
Dr	Cr
Prepayment from Customer	Sales of Goods (reversal)
	Current VAT Payables (reversal)
Sales of Goods (Return of Goods)	Prepayment from Customer

> ***Real-World Explanation:***
> *Zara returns 500 defective hangers (Value: AED 200 + AED 10 VAT).*
>
> - ***Dr Sales of Goods****: AED 200 (Reversing the income).*
> - ***Dr Current VAT Payables****: AED 10 (We no longer owe this tax).*
> - ***Cr Prepayment from Customer / Trade Receivables****: AED 210 (Reducing what Zara owes us, or creating a credit balance for them).*

5.5 Mutual Settlement (AP/AR Netting) — GL Mapping

```
Mutual Settlement:
  Dr: Current Trade Payables (or Receivables)
  Cr: Current Trade Receivables (or Payables)
  (Net off the balances between customer/supplier who is both)
```

> ***Real-World Explanation:***
> *Logistics Plus LLC provides trucking services for us, and they recently billed us AED 10,000 (They are a Supplier in AP). However, they also buy packing materials from us and owe us AED 4,000 (They are a Customer in AR).*
>
> *We perform a mutual settlement of AED 4,000 to net the accounts:*
>
> - ***Dr Current Trade Payables****: AED 4,000 (Reduces our AP liability*

> *to them down to 6k).*
> - ***Cr Current Trade Receivables**: AED 4,000 (Reduces their AR balance to 0).*

> *Result: No cash moves through the bank, but both ledgers are instantly reconciled.*